Certified Reference Material

BAM
Bundesanstalt für Materialforschung und -prüfung

BAM-D001

Particle Size Distribution by Laser Diffraction Methods according to ISO 13320

Certified Values

Specific Particle Diameter Corresponding to the Cumulative Undersize Volume Distribution Q3	Equivalent Spherical Diameter [1] in µm	Uncertainty *U* [2] in µm
d_{10}	7.02	0.25
d_{50}	12.48	0.21
d_{90}	20.8	1.1

[1] The certified value is the weighted mean of 13 laboratory means which participated in the interlaboratory comparison for certification according to ISO 13320:2009.

[2] Estimated expanded uncertainty *U* with a coverage factor of *k* = 2, corresponding to a level of confidence of about 95 %, as defined in the Guide to the Expression of Uncertainty in Measurement (ISO/IEC Guide 98-3:2008).

Values for Information

Refractive index n_p [3]	2.645
Imaginary part k_p of particles' refractive index (absorption) [3]	0.1
Density ρ in g/ cm^3	3.205

[3] Wavelength: 633 nm

This certificate is valid for a period of five years beginning with the dispatch of the reference material from BAM.

Date of Dispatch:

Material Description

A unit of BAM-D001 consists of a single glass bottle containing at least 10 g of hexagonal silicon carbide powder.

Intended Use

The certified reference material BAM-D001 is intended for checking instruments and the verification of analytical results determined by laser diffraction methods according to ISO 13320.

Handling

Wet dispersion

The powder has to be dispersed in a liquid. Therefore a 0.003 mol/L tetrasodiumdiphosphate solution should be used. 0.798 g $Na_4P_2O_7$ have to be dissolved in 1 L distilled water to produce this solution.

Ultrasonic treatment

Approximately 0.3 g of the powder has to be stirred in 80 ml of the 0.003 mol/L $Na_4P_2O_7$-solution. The suspension has to be treated by ultrasound for at least three minutes. Preferably a high performing external ultrasound disintegrator should be used.

After ultrasonic treatment the suspension has to be cooled at room temperature in a cold water quench and must be stirred by a magnetic agitator until the measurement will be started.

Measurement

0.003 mol/L $Na_4P_2O_7$ solution has to be given into the dispersion unit of the measuring device. Then a background measurement has to be performed.

After background measurement a small amount of the cooled dispersed suspension has to be taken by a pipette and added to the dispersion unit. The sample concentration of the solution has to be selected according to the recommendation of the measuring device manufacturer. The suspension has to be agitated during background and sample measurements.

It is important to avoid bubbles in the suspension during the measurements.

Mie theory should be used for the calculation of the particle size distribution. The following optical parameters have to be used for the calculation procedure:

refractive index n_p :	2.645
imaginary part of the refractive index k_p (absorption):	0.1
refractive index n_m of the dispersion liquid (water):	1.33

If possible, a calculation model regarding the irregular shape of the particles should be used.

The density ρ of the material, measured by Helium pycnometry (DIN 66137-2:2004), is 3.205 g/cm^3.

Comparison of a measured result with the certified values

The verification of results by use of the reference material BAM-D001 should be performed according to Application Note 1 (Thomas Linsinger, EC, IRMM, 2005). Therefore a comparison of the difference of the measured value and the certified value (Δ_m) with the expanded uncertainty (U_Δ) should be used as follows:
At first the absolute value of difference of the measured value and the certified value has to be calculated:

$$\Delta_m = \left| d_m - d_{CRM} \right|$$

d_m……measured value

d_{CRM}……certified value

Secondly, the uncertainty of Δ_m has to be calculated:

$$u_\Delta = \sqrt{u_m^2 + u_{CRM}^2}$$

u_Δ ……combined uncertainty of the measured value and the certified value

u_m……uncertainty of the measured value

u_{CRM}……uncertainty of the certified value

The uncertainty of the certified value is calculated by dividing the expanded uncertainty *U,* shown at the certificate, by the factor $k = 2$.

The uncertainty of the measured value can be expressed approximately by use of the reproducibility standard deviation which was determined by the interlaboratory comparison for the certification (for data base cp. attachment 3 of the certification report BAM-D001):

d_{10}-value: $u_m = 0.4$ µm

d_{50}-value: $u_m = 0.28$ µm

d_{90}-value: $u_m = 1.7$ µm

The calculated combined uncertainty has to be multiplied by the factor $k = 2$. The result correlates to a confidence level of approx. 95 %.

$U_\Delta = 2 \times u_\Delta$

In a final step Δ_m has to be compared with U_Δ to evaluate the performance of the measurement. There is no significant difference between the measured value and the certified value if $\Delta_m \leq U_\Delta$.

Transport und Storage

The material has to be stored in a dry atmosphere.

Participating Laboratories

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Nabaltec AG, Schwandorf, Germany

Retsch Technology GmbH, Haan, Germany

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References

Zertifizierungsbericht BAM-D001 (P. Kuchenbecker, M. Gemeinert, BAM, February 2012)
www.bam.de

Guidelines for the Production of BAM Reference Materials (BAM, 2010)
www.bam.de/pdf/service/referenzmaterialien/bam_rm_leitfaden.pdf

ISO 13320:2009: Particle size analysis — Laser diffraction methods

ISO Guide 35: 2006: Reference Materials – General and statistical principles for Certification

ISO/IEC Guide 98-3:2008: Uncertainty of measurement – Part 3:
Guide to the expression of uncertainty in measurement (GUM)

Application Note 1 (T. Linsinger, EC, IRMM, 2005)
http://www.bam.de/de/fachthemen/referenzmaterialien/referenzmaterialien_medien/erm_application_note_1_ge.pdf

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